

Home Network Security Assessment & Baseline Mapping

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Environment: Home Network (ISP-Managed Router – Spectrum SAX1V1K)

Assessment Type: Defensive Security Assessment / Network Baseline Mapping

Summary

This project documents and validates the security posture of a residential network by establishing a complete network baseline, identifying active devices, verifying perimeter exposure, and assessing wireless and router security controls.

Due to the use of an ISP-managed router, configuration and monitoring capabilities were limited. However, a thorough assessment was still completed using host discovery, port scanning, service enumeration, and validation testing.

Outcome

There were no externally exposed services detected. All internal devices were fully inventoried. The guest Wi-Fi isolation was validated. Confirmed UPnp was disabled and clear limitations documented.

Scope & Objectives

Scope

- Local Area Network (192.168.1.0/24)
- ISP router (LAN & WAN perspectives)
- Wireless configuration and guest network
- Device inventory and baseline creation

Objectives

- Identify all active devices (IP, MAC, vendor)
- Document open ports and services
- Validate router exposure and UPnP status
- Establish a defensible baseline for future comparisons

Network Mapping & Baseline Inventory

Objective

Document all devices, IP addresses, MAC addresses, manufacturers, and exposed services on the home network.

All discovery was performed from a Kali Linux VM connected to the same LAN.

Host Discovery – ARP Scan

ARP scanning was used to identify all Layer-2 reachable devices, including those that may not respond to ICMP.

```
sudo apt install -y arp-scan
```

```
sudo arp-scan --localnet --interface eth1 | tee  
report/arp_scan.txt
```

Result:

- Enumerated all active LAN devices
- Identified router, desktops, laptops, consoles, and mobile devices
- Captured IP ↔ MAC relationships for baseline inventory

arp-scan output:

```
(chontele@Kali)-[~/home-network-segmentation/report]
$ sudo arp-scan --localnet --interface=eth1 | tee report/arp_scan.txt
tee: report/arp_scan.txt: No such file or directory
Interface: eth1, type: EN10MB, MAC: 08:00:27:62:b2:e3, IPv4: 192.168.1.118
Starting arp-scan 1.10.0 with 256 hosts (https://github.com/royhills/arp-scan)
192.168.1.74: [REDACTED] ASKEY COMPUTER CORP
192.168.1.3c: [REDACTED] Liteon Technology Corporation
192.168.1.80: [REDACTED] (Unknown)
192.168.1.54: [REDACTED] (Unknown)
192.168.1.84: [REDACTED] Sony Interactive Entertainment Inc.
192.168.1.fc: [REDACTED] (Unknown)
192.168.1.98: [REDACTED] Samsung Electronics Co.,Ltd
192.168.1.de: [REDACTED] (Unknown: locally administered)

8 packets received by filter, 0 packets dropped by kernel
Ending arp-scan 1.10.0: 256 hosts scanned in 2.192 seconds (116.79 hosts/sec). 8 responded
```

Network Mapping – Ping Sweep

A lightweight Nmap ping sweep was performed to confirm live hosts and hostnames.

```
sudo nmap -sn 192.168.1.0/24 -oN report/nmap_ping_sweep.txt
```

Purpose:

- Validate host availability
- Identify device naming patterns
- Cross-reference ARP results

Ping Sweep output:

```
(chontele@Kali)-[~/home-network-segmentation]
$ sudo nmap -sn 192.168.1.0/24 -oN report/nmap_ping_scan.txt
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-22 09:26 EST
Nmap scan report for [REDACTED]
Host is up (0.49s latency).
Nmap scan report for [REDACTED] ([REDACTED])
Host is up (0.0054s latency).
Nmap scan report for [REDACTED]
Host is up (0.32s latency).
Nmap scan report for [REDACTED]
Host is up (0.32s latency).
Nmap scan report for [REDACTED] ([REDACTED])
```

Full Internal Network Scan – Nmap Analysis

A comprehensive Nmap scan was performed across the internal subnet (192.168.1.0/24) to identify **open ports, running services, and potential OS fingerprints**:

```
sudo nmap -sS -sV -O 192.168.1.0/24 -oA report/nmap_full_scan
```

Summary of Findings:

- **Device Discovery:** All hosts identified in previous ARP and ping sweeps were confirmed.
- **Port Status:**
 - Most ports on LAN hosts are **closed** or **filtered**, reflecting strong internal device security.
 - The router shows only port 53 (DNS) open internally; all other ports are closed or filtered.
- **OS Detection:**
 - OS fingerprinting was **unreliable** for some hosts due to the absence of open ports. This is expected in a secured home environment and is a positive indicator of limited exposure.
- **Warnings Observed:**
 - **RTTVAR has grown...** messages indicate Nmap's adaptive timing in response to filtered or non-responsive ports. This is normal in environments with stateful firewalls.
 - **Retransmission cap hit** warnings occur when the router does not respond to repeated probes, demonstrating proper network hardening against unsolicited scans.
 - OS detection limitations are not a vulnerability; they are a consequence of the network being intentionally restrictive.

Interpretation:

The scan demonstrates a well-secured internal network with no unnecessary service exposure. Host devices and the router are resistant to unsolicited probing, and OS fingerprinting limitations highlight the effectiveness of these defensive controls. These results confirm the internal network baseline established through ARP scans and ping sweeps.

Nmap_full_scan.txt results:

```
GNU nano 8.4 nmap_full_scan.nmap
# Nmap 7.95 scan initiated Sat Nov 22 09:30:51 2025 as: /usr/lib/nmap/nmap -sS -sV -O -oA repo>
RTTVAR has grown to over 2.3 seconds, decreasing to 2.0
RTTVAR has grown to over 2.3 seconds, decreasing to 2.0
RTTVAR has grown to over 2.3 seconds, decreasing to 2.0
RTTVAR has grown to over 2.3 seconds, decreasing to 2.0
RTTVAR has grown to over 2.3 seconds, decreasing to 2.0
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RTTVAR has grown to over 2.3 seconds, decreasing to 2.0
RTTVAR has grown to over 2.3 seconds, decreasing to 2.0
RTTVAR has grown to over 2.3 seconds, decreasing to 2.0
Warning: giving up on port because retransmission cap hit (10).
Nmap scan report for 
Host is up (0.028s latency).
All 1000 scanned ports on  are in ignored states.
Not shown: 721 closed tcp ports (reset), 279 filtered tcp ports (no-response)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 clos
OS fingerprint not ideal because: Missing an open TCP port so results incomplete
No OS matches for host

Nmap scan report for 
Host is up (0.0044s latency).
Not shown: 883 closed tcp ports (reset), 114 filtered tcp ports (no-response)
PORT      STATE SERVICE  VERSION
open  domain          DNS
```

Router-Focused Port & Service Scan (Internal)

The router was scanned from the LAN to identify management or discovery services.

```
sudo nmap -p 1-65535 -sV -T4 -Pn 192.168.1.1 -oN
report/router_ports.txt
```

Result:

- No unnecessary LAN-side services exposed
- Router management locked to internal interface

Router port scan results:

```
(chontele@Kali)-[~/home-network-segmentation]
$ sudo nmap -p 1-65535 -sV -T4 -Pn [REDACTED] -oN report/router_ports.txt

[sudo] password for chontele:
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-23 14:29 EST
Nmap scan report for [REDACTED]
Host is up (0.0032s latency).
Not shown: 64512 closed tcp ports (reset), 1020 filtered tcp ports (no-response)
PORT      STATE SERVICE  VERSION
53/tcp    open  domain   Cloudflare public DNS
80/tcp    open  http     lighttpd 1.4.59
443/tcp   open  ssl/http lighttpd 1.4.59
MAC Address: [REDACTED]
```

UPnP Validation

UPnP was explicitly checked to ensure no automatic port forwarding was active.

```
sudo nmap -p 1900 --script upnp-info 192.168.1.1 -oN
report/upnp_check.txt
```

Result:

- UPnP service closed
- No UPnP discovery responses

UPnP results:

```
(chontele@Kali)-[~/home-network-segmentation]
$ sudo nmap -p 1900 --script upnp-info [REDACTED] -oN report/upnp_check.txt
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-16 12:29 EST
Nmap scan report for [REDACTED]
Host is up (0.0046s latency).

PORT      STATE SERVICE
1900/tcp   closed upnp
MAC Address: [REDACTED]

Nmap done: 1 IP address (1 host up) scanned in 0.34 seconds
```


Local Neighbor Cache Validation

Local ARP and neighbor tables were reviewed to confirm consistency.

`ip neigh show`

Purpose:

- Validate ARP scan accuracy
- Detect transient or sleeping devices

IP neigh show results:

```
(chontelee@Kali)-[~/home-network-segmentation/report]
$ ip neigh show
[REDACTED] dev eth0 lladdr [REDACTED] STALE
[REDACTED] dev eth1 lladdr [REDACTED] REACHABLE
[REDACTED] dev eth1 lladdr [REDACTED] STALE
[REDACTED] dev eth0 lladdr [REDACTED] STALE
[REDACTED] dev eth1 lladdr [REDACTED] STALE
[REDACTED] dev eth0 lladdr [REDACTED] STALE
[REDACTED] eth1 lladdr [REDACTED] router STALE
[REDACTED] dev eth1 lladdr [REDACTED] router STALE
[REDACTED] dev eth1 lladdr [REDACTED] STALE
```


Wireless Security Assessment

Observed Controls:

- WPA2/WPA3-Personal enabled
- SSID broadcast enabled
- Guest network enabled with isolation
- WPS not exposed via UI

Wi-fi router settings:

Router Name Router	Rename >
Model [REDACTED]	
Serial Number [REDACTED]	
Public IPv4 [REDACTED]	
Public IPv6 [REDACTED]	

Advanced Settings

Set Up Smart Device on 2.4 GHz >

Port Forwarding & IP Reservations >

DNS Server >

UPnP ⓘ ☐

Validation & Attack Simulation (Non-Intrusive)

Guest Network Isolation Test

- Connected a device to guest Wi-Fi
- Attempted to reach LAN IPs

ping 192.168.1.71

Result:

- No response → isolation confirmed

Guest ping results:

```
(chontele@Kali)-[~/home-network-segmentation]
$ ping 192.168.1.71 -t4
PING 192.168.1.71 (192.168.1.71) 56(84) bytes of data.
^C
— 192.168.1.71 ping statistics —
36 packets transmitted, 0 received, 100% packet loss, time 35870ms
```

Limitations

- ISP-managed router prevents:
 - VLAN creation
 - Firewall rule tuning
 - Syslog or traffic logging
- No Wi-Fi adapter available for packet capture
- Some devices were offline during scanning

These limitations were documented and did not invalidate results.

Risk Summary

Area	Risk	Notes
External exposure	Low	No open WAN ports
Internal segmentation	Medium	Flat LAN
Wireless security	Low	WPA2/WPA3 enforced
Monitoring	Medium	No logs available

Recommendations

- Upgrade to a user-managed router/firewall (pfSense, Ubiquiti, OPNSense)
- Segment IoT devices into separate networks
- Enable logging and alerting
- Re-run baseline scans quarterly

Conclusion

This assessment successfully established a defensible network baseline, validated router and wireless security controls, and documented real-world limitations imposed by ISP hardware.

The project demonstrates hands-on defensive security skills, practical tool usage, and professional documentation suitable for SOC, GRC, or IT security roles.